



Kenya Certificate of Secondary Education (K.C.S.E).

443/1

AGRICULTURE

PAPER 1

TIME: 2 HOURS

MARKING SCHEME

SECTION A

1. (a) (i) Plantation farming – i.e the growing of only one type of crop on large scale. (1x1=1mk)
(ii) Large scale farming is the growing of crops and rearing of animals in large numbers for commercial purposes. (1x1=1mk)
- (b) Arid and semi arid areas do not support crops, rather suitable for natural grasses and legume plants.
Ranching requires large or wide grazing land which is available in marginal areas. (Any 1 x 1 = 1mk)
2. (a) (i) Some implements or tools cannot produce a fine tilth required by small planting materials (1x1=1mk) OR ANY OTHER ALTERNATIVE RESPONSE
(ii) Cost of the tool is required for budgeting and this depends on the financial status of the farmer, choose a tool the farmer can afford. (1x1=1mk)
OR ANY OTHER ALTERNATIVE RESPONSE
- (b) - Ridging
- Rolling
- Leveling (2x ½ = 1mk)
3. - Temperature
- Wind
- Rainfall (Any 2 x ½ = 1mk)
4. - Soils with rough texture have a low water holding capacity crops growing in them may lack adequate moisture (crops can wilt)
- Rough soils get easily leached and they lose nutrients easily
- Rough soils can be easily washed away hence lose of soil fertility. (Any 2 x 1 = 2mks)
5. - Increases the water holding capacity of the soil
- It improves soil fertility by releasing a wide range of nutrients into the soil.
- It improves soil structure / Birds soil particles
- Buffers soil PH
- Provide food for the soil organisms (Any 1x1=1mk)
6. - Respiration
- Combustion
- Decomposition. (Any 2x1=2mks)
7. - Marcotting / Aerial layering
- Tip layering
- Trench layering
- Compound / serpentine layering
8. Topping is the cutting of the stemy, fibrous remains of pasture after direct grazing by the use of a sharp Panga. While top dressing is the application of Nitrogenous fertilizer in the field of established crops. (1x1=1mk)



9.
 - Too much nitrogen in early stages of growth
 - Irregular or frequent watering
 - Calcium deficiency compounds in the soil

(Any 2x1=2mks)
10.
 - To ease population pressure from over populated areas.
 - To increase agricultural production through making better use of uninhabited or idle land.
 - To create employment – when people are given land to do farming they get self employment.
 - To control tse tse flies – when land is cultivated it became a barrier to the movement of tsetse flies.

(Any 2x1=2mks)
11. (a) Decision making
 - Gathering information
 - Compare standards of one's enterprise with the set standards
 - Detects weaknesses and constraints and find ways and means of overcoming them.
 - Keep farm records up-to-date and use them in the day – to – day running of the farm
 - Implements from decisions and taking responsibility.

(Any 4 x ½ = 2mks)
- (b)
 - Stalk borers
 - Nematodes

(2x1=2mks)
12.
 - Use healthy plantating materials
 - Practice field hygiene e.g burning diseased crop residues.
 - Proper seed preparation before planting
 - Proper spacing in the nursery and seedbed
 - Proper drying of cereals and pulses to recommended moisture before storage.
 - Use disease resistant crop varieties
 - Crop rotation

(Any 2 x ½ = 1mk)
13.
 - Invoice
 - Statements
 - Receipts

(Any 2x1=2mks)
14.
 - Threshing / shelling
 - Drying
 - Cleaning
 - Sorting and grading
 - Dusting

(Any 2x1=2mks)
15. Environmental factors that affect the effectiveness of herbicides.
 - wind
 - rainfall
 - soil
 - temperature
 - light

(2mks)
16. ways of improving land as a factor of production
 - irrigation
 - Pests and disease control
 - fencing
 - Fertilize application
 - Paddockking

(4x ½ =2mks)
17. Ways of improving labour in the farm
 - training



- farm mechanization
- giving incentives
- labour supervision
- improving terms and conditions of service.

(4x ½ = 2mks)

18. Name two methods of harvesting agro forestry trees.

(1mk)

- Coppicing
- Lopping
- Pollarding

(2 x ½ = 1mk)

SECTION B

19. (a) Cost of fertilizer

When 3 bags were used

$$= 3 \text{ bags} \times \text{Ksh } 300 = \text{Ksh } 900$$

(1 x ½ = ½ mk)

When 4 bags were used

$$= 4 \text{ bags} \times \text{Ksh } 300 = \text{Ksh } 1200$$

(1 x ½ = ½ mk)

(b) Marginal revenue at level 5 and 6 can be found after completing the question table as follows.

DAP fert. In (10kgbag)	Maize yield in (20kgbag)	TR (Ksh)	TC(Ksh)	MR	MC
0	10.5	2,100	-	-	
1	20.5	4,100	300	2000	
2	42.5	8,500	600	4400	
3	58.5	11,700	900	3200	
4	60.0	12,000	1,200		
5	60.5	12,700	1,500		
6	58.5	11,700	1,800		
7	56	7,200	2,100		

$$MR_2 = 8,500 - 4100 = 4400 \checkmark 1mk$$

$$MR_3 = 11,700 - 8500 = 3200 \checkmark 1mk$$

20. (a) Chitting / sprouting

(1x1=1mk)

(b) P – Rose end

(1x1=1mk)

Q – Heel end

(1x1=1mk)

- (c) - Crop originating from vegetative material mature faster
- Crops show uniformity
- Can be used because Irish potatoes do not produce seeds.

(Any 1 x 1 = 1mk)

21. (a) NPK (25:20:15)

25 – 25kg or %

Nitrogen

20 – 20kg or %

Phosphorus Pentaoxide

15-15kg or %

Potassium oxide / chloride

(b)

Fertilizer	Class	Reason
NPK (25:20:15)	Complete compound fertilizer ✓ 1mk	Contains all the three primary macro-element. ✓ 1mk
Urea (46:18:0)	Incomplete compound fertilizer ✓ 1mk	Contains only two of the primary macro-nutrients ✓ 1mk

22. (a) Sorghum shoot fly / shoot fly

(1x1=1mk)

(b) Dead shoot of the sorghum crop

- (c) - Early planting
- Closed season
- Application of insecticides



23. (a) (i) Broad based terrace (1x1=1mk)
 (ii) Bund terrace (1x1=1mk)
 (b) (i) Top channel = 1.5m (1x ½ = ½ mk)
 (ii) Bottom channel = 90cm (1 x ½ = ½ mk)

SECTION C

24. (a) Reasons for controlling weeds.
 - Avoid weeds which compete with crop nutrients, space, light and soil moisture.
 - Remove parasite weeds like the witch weed.
 - Improve the quality of produce which can be reduced by presence of weed seeds, and bad weed smells
 - Avoid farm poisoning by poisonous weeds like thorn apple.
 - To control some pests which can be hosted by some weeds.
 - To allow better crop germination by removing weeds that are allelopathic.
 - Open irrigation channels
 - improves the quality of pastures
 - Ease work by killing irritating weeds.
 (Any 5 correctly explained x 2 = 10mks)
- (b) (i) Cut the crop when 50% of the plant has flowered. Spread out the cut material on the ground evenly to dry for 2-3 days
 (ii) Spread out the cut material on the ground evenly to dry for 2-3 days.
 (iii) The hay is windrowed and then baled.
 (iv) The hay bales are then stored in a shed out of reach by rain water and sunshine.
 (Step (i) – (ii) correctly followed = 4 mks)
- (c) - Source to firewood / food fuel
 - Agro forestry trees or products can be sold to earn income.
 - Trees protect soil from strong rains, sun and wind and consequently reduce soil erosion.
 - Sales time wasted to look for firewood
 - Makes farms beautiful
 - Source of construction materials
 - Marks farm boundaries
 - Provide shade to crops and livestock
 - Can be eaten as food
 - Can be used as livestock feeds
 (Any 6 correctly explained = 6mks)
25. (a) (i) - Dig deeply to remove all weeds
 - Carry out secondary to produce fine tilth
 - Level the ground to ensure even germination
 (3x1=3mks)
- (ii) - Plant at the onset of rains
 - Plant by broadcasting, if planted in rows, the furrows should be 30-33 cm apart.
 - Plant shallowly not more than 2.5cm deep.
 - Plant using phosphate fertilizers.
 (4x1=4mks)
- (iii) - Hand knives are used for cutting heads
 - Harvest when heads have dried
 - Harvest into bags, dry, thresh and winnow
 (3x1=3mks)
- (c) (i) Biotic factors



Usefulness	Harmful effects
-Some help in the decomposition of organic matter -Predators reduce pest population -Nitrogen fixing Bacteria improve soil fertility -Pollination carry out pollination (Any 2x ½ =1mk)	-Pests destroy crops -Parasites suck blood and transmit livestock diseases -Pathogens cause diseases -Predators kill livestock (Any 2x ½ =1mk)

(ii)

How government policy encourage agriculture	How the government discourages agriculture
-Gives subsidies or make part payment for the farmers -Boost the control of pests, diseases and parasites -Conserves environment -Give financial support to farmers (Any 2x ½ =1mk)	-Heavy taxation -Prohibits production of some good e.g. local beer -Use of quote system e.g. in coffee production. -Predators kill livestock (Any 2x ½ =1mk)

26. (a) (i) - Cattle bomas may have high levels of soil nutrients which is not a true picture of that soil fertility.
 - Animals might have trampled soil to destroy soil structure. (2x1=2mks)
- (ii) - Along water ways soil erosion has taken place
 - Along water ways soil deposition takes place (2x1=2mks)
- (b) - Near source of water to some time and labour
 - Site previously not planted crops of the same family to control pests, diseases and weeds.
 - Away from natural sheds
 - Seared from thieves and animals to avoid destruction
 - Gently sloping ground for easy drainage
 - Accessible place for easy transportation
 - Fertile soil for better provision of plant nutrients. (Any 4x1=4mks)
- Reject**
- Availability of water
 - Security
 - Topography
 - Accessibility
 - Soil fertility
- (c) (i) Clear vegetation from the soil sampling spots and make vertical cut of 15-25cm for crop level and 5cm for pasture land.
 (ii) Take a slice from the vertical cut using a spade or soil anger.
 (iii) Put the sampled soil in a clean polythene bag or any suitable container.
 (iv) Repeat steps (i), (ii) and (iii) in different pairs of the field 15-20 spots.
 (v) Mix thoroughly the soil sample and mix thoroughly, dry and crush
 (vi) Send a sub-sample from the mixture and to the laboratory (6steps x 1 = 6mks)
- Note:** Steps should be strictly followed if otherwise reject.
- (d) - Can be used to secure loans
 - Help in making management decisions
 - Help to determine whether the farm is making profits
 - Help in farm budgets i.e to draw estimates for future income and expenditure.
 - For evaluating assets and liabilities
 - Helps farmers to estimate the amount of tax that can be charged so that the farm is fairly taxed. (6x1=6mks)